Young Hyun Kim

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EDUCATION *(Sejong University, Seoul, Republic of Korea)*

M.S. in Integrative Biological Sciences and Industry Aug 2026
Thesis: Functional characterization of hiPSC-derived chamber-specific cardiac organoids using calcium-imaging and contractility analysis.

B.S. in Integrative Biological Sciences and Industry Aug 2023

RESEARCH EXPERIENCE

Graduate Research Assistant — Biological Systems Engineering Laboratory Sep 2023 – Present
Dept. of Integrative Bioscience & Technology, Sejong University, Seoul, Korea
PI: Prof. Keel Yong Lee

Thesis project — Chamber-specific cardiac organoid platform

- **Fabrication:** Differentiated hiPSCs into chamber-specific (ventricular and atrial) cardiomyocytes via temporal dual modulation of canonical Wnt signaling, using retinoic acid (1 µM) to direct atrial versus ventricular specification, and fabricated 2D/3D cardiac organoids, achieving ~97.5% cTNT+ ventricular purity by flow cytometry.
- **Structural characterization:** Established a chamber-identity characterization workflow combining qRT-PCR (NR2F2, IRX4), immunofluorescence, and flow cytometry (TNNT2); built a MATLAB Organoid Diameter Analyzer validated against manual measurement by Bland–Altman analysis.
- **Functional characterization:** Designed, built, and validated iCAMAnalyzer, a unified Python-based tool integrating calcium-imaging and motion-tracking analysis of cardiac organoids ($\Delta F/F_0$ normalization, BPM, amplitude, CTD60/90, τ, Vmax/Vrelax, activation mapping, conduction-velocity estimation, contraction magnitude/duration), achieving ~7% error against Python-simulated ground-truth data; supports simultaneous multi-organoid (9-well) imaging with automated ROI detection.

Additional Projects & Lab Responsibilities

- Currently developing a spinner-flask (stirred-suspension) bioreactor protocol to scale up hiPSC-cardiomyocyte production, supporting higher-throughput organoid fabrication. [in progress]
- **Parallel project —** Engineered 3D skeletal muscle tissue on custom, 3D-printed micropillar scaffolds — designed using COMSOL Multiphysics simulations of microfluidic fluid velocity and electric-potential fields — and assessed contractile function and calcium-signal propagation.
- **Collaborative contribution:** Provided hiPSC-derived cardiomyocytes as a shared resource to an inter-lab collaboration with Dr. Chun Jeih Ryu's cancer research group, supporting a co-authored study on CD147/CD98hc-targeted hepatocellular carcinoma therapy (Cancer Research Communications, under review).
- Manage ongoing laboratory operations, including budget/purchasing, reagent and inventory tracking, new-member onboarding, and lab meeting scheduling.

Undergraduate Research Assistant Oct 2022 – Aug 2023
Dept. of Integrative Bioscience & Technology, Sejong University, Seoul, Korea
PI: Prof. Keel Yong Lee

- Independently developed a protocol for fabricating iPSC-derived cardiac organoids, as no established protocol existed in the lab at the time — laying the groundwork for the lab's ongoing organoid research.
- Fabricated 3D skeletal muscle tissue constructs from C2C12 myoblasts and applied cyclic mechanical stretch to assess its effect on tissue maturation.
- **Founding member of the laboratory:** led initial equipment procurement and setup, and established experimental protocols (SOPs) and safety regulations (safety oversight since transitioned to another lab member).

PUBLICATIONS

1. **Kim, Y. H.**, Son, Y. H., Choi, Y., Kim, M. S., Park, S. J., & Lee, K. Y. (2026). Advanced in vitro cardiac models for drug evaluation: Integration of organoids, engineered tissues, and microphysiological systems. *Microsystems & Nanoengineering*, 12(1), 162. **(First author)**
2. Kim, M. S., **Kim, Y. H.**, Choi, Y., & Lee, K. Y. (2025). Biohybrid actuators in robotics: recent trends and future perspectives of skeletal and cardiac muscle integration. *npj Robotics*, 3(1), 37. (Second author)
3. Kim, Y. S., Lee, M., Cheon, Y. J., Choi, H. S., Choi, B. J., **Kim, Y. H.**, Lee, K. Y., & Ryu, C. J. (2026). Targeting CD147 or CD98hc is sufficient to collapse the CD147–CD98hc–MCT1/MCT4 module and restrain hepatocellular carcinoma growth. *Cancer Research Communications*. (Contributed Resources, Visualization)

CONFERENCE PRESENTATIONS

- Poster, Korea BioChip Society (Nov 2025, Jeju) — Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis
- Poster, 2025 Summer Symposium on Biological Convergence Engineering (Sep 2025, Sejong University) — Chamber-Specific Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis

AWARDS

Grand Prize — 4th College of Life Sciences Academic Competition May 2016

Sejong University, Seoul, Korea — 5-member team

- Awarded top prize for a food-engineering project, "Benzopyrene Reduction Using Enzymes of Alteromonas naphthalenivorans," among student teams at a college-wide academic competition.

INSTRUMENTATION & ENGINEERING PROJECTS

Bio-Synchronous 3D Tissue Electromechanical Stimulator 2025

- Designed a modular electromechanical stimulation platform for 3D cardiac tissues: electrode assembly, humidity control, integrated webcam imaging (replacing a transparent-window design), and a static-mold add-on for control samples. [in progress]

TEACHING & MENTORING EXPERIENCE *(all at Sejong University, Seoul, Korea)*

Undergraduate Research Mentor — SURF Program Jul 2026 – Aug 2026

- Coached 4 undergraduate students during a month-long, independent SURF research cohort, guiding the group through experimental design and independent data generation in the lab.

Teaching Assistant — Biochemistry Laboratory Jul 2023 – Dec 2024

- Guided ~120 undergraduate students (three lab sections of ~40) through plasmid transformation, protein expression and purification, and Western blotting; prepared course materials and ensured protocol compliance.

Teaching Assistant — Microbiology Laboratory Jan 2024 – Jun 2024

- Instructed 40 undergraduate students in bacterial culture, Gram staining, morphological observation, and basic biochemical testing.

Peer Research Mentor — DoLearn-DoRun Program Sep 2023 – Dec 2023

- Mentored 2 undergraduate students for one semester as part of a graduate-led peer research team: authored the team's research proposal and directed all experimental work.

Science Education Volunteer — PBL Creative Sharing Volunteer Competition Sep 2016 – Dec 2016

- Led hands-on science experiments for 20 elementary school students over 10 sessions as a team member in a science-education volunteering program.

TECHNICAL SKILLS

Cell & Tissue Engineering: hiPSC culture and chamber-specific cardiomyocyte differentiation; 3D cardiac organoid fabrication; 3D skeletal muscle tissue engineering.

Assays & Imaging: flow cytometry (cTNT/TNNT2), qRT-PCR (NR2F2, IRX4), immunofluorescence, calcium-transient imaging, contractility analysis.

Computational & Programming: MATLAB (App Designer), Python (Streamlit, NumPy/SciPy), ImageJ, GraphPad Prism; custom analysis-tool and pipeline development.

Simulation & Fabrication: COMSOL Multiphysics (fluid velocity, electric potential, hyperelastic deformation); 3D modeling (Rhino) and printing.

Scientific Illustration: Inkscape, Adobe Illustrator — hand-drawn all schematic figures for both first- and second-author peer-reviewed articles.

LANGUAGES

Korean: Native
English: Proficient

REFERENCES

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